Akka vs. Temporal

A comparison for teams building agentic AI

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Temporal runs your workflows; Akka runs your whole agentic system — and guarantees it. Temporal is excellent durable execution for one layer: orchestration. An agentic system also needs agents, memory, streaming, APIs, and governance. With Temporal you source, integrate, and operate the agent framework, memory, API layer, and governance tools yourself. Streaming agent output with Workflow Streams is in public preview. The Akka Agentic AI Platform delivers them as one system that guarantees resilience and scalability.

99.99%	99.9999%	4ms	5B
TEMPORAL SLA	AKKA SLA	AKKA STATE READS	TOKENS / SEC BENCHMARK

DIMENSION	TEMPORAL	AKKA
What it is	A durable-execution / workflow-orchestration engine	The Akka Agentic AI Platform — guarantees resilience and scalability
Scope	Orchestration, plus Workflow Streams for agent output (public preview); agent frameworks, memory, APIs, and governance are sourced and integrated by the customer	Orchestration, agents, memory, streaming, APIs, observability, and governance on one runtime
Availability SLA	99.9% standard; 99.99% High Availability / multi-region	99.9999% — entire platform, backed by indemnities
RTO / RPO	~20-minute RTO; sub-1-minute RPO (HA namespaces)	Sub-1-minute RTO; zero-byte RPO
AI agents	SDK integration (e.g., OpenAI Agents SDK); not native	Native agents with tools, handoffs, guardrails, interaction logging
Memory	Workflow state in the Event History; agent memory via a third-party integration (e.g., xMemory)	Durable in-memory, 4ms reads / sub-10ms writes
Governance / EU AI Act	Infrastructure compliance; no first-party AI policy enforcement, explainability, classification, or regulation mapping; partner guardrails (Tenuo, OpenBox) for Temporal workflows only	Aspect-woven runtime enforcement + full pre-production governance
Programming model	Workflow code must be strictly deterministic	Natural Java/Scala; durability handled by the runtime
Cost model	Per-action billing ($50→$25 per million) + separate infra for everything else	Shared compute; up to 90% lower infrastructure for the same workload
Certifications	SOC 2 Type II, GDPR, HIPAA	19 standards (SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF)

Temporal Is One Layer; Akka Is the Full Stack

Temporal solves durable execution of long-running workflows, and it solves it well. But an agentic AI system needs far more than orchestration, and Temporal leaves the rest to you. You select, wire, and operate the memory layer, the data-streaming tier, the API layer, the agent framework, and the partner governance tools around the orchestration core, and you own every failure between them.

Everything an agentic system needs runs in one Akka runtime, so a system that compiles is ready to run in production. The Akka SDK models the entire agentic system in one place — agents, tools, durable memory, streaming, APIs, and governance — and the platform

generates it, tests it, and runs it as one certified service.

Capability	Temporal	Akka
Workflow orchestration	Yes	Yes
Native AI agents (tools, handoffs, streaming)	SDK integration only	Built in
Durable memory	Workflow state in the Event History; agent memory via third-party integration	Built in, 4ms / sub-10ms
Real-time streaming	Workflow Streams for agent output to a UI (public preview); data pipelines provisioned separately	Built in, backpressured, petabyte-scale
HTTP / gRPC API layer	None	Built in
Governance / policy enforcement	Partner integrations (Tenuo, OpenBox); Temporal workflows only	Inline, runtime-embedded
Pre-production governance	None	Classification, sign-offs, sealed posture

Availability and Disaster Recovery

Temporal Cloud has materially strengthened its HA/DR: High Availability and multi-region namespaces publish a 99.99% availability SLA (99.9% standard) with sub-1-minute RPO and ~20-minute RTO via automatic failover. That is real. Two gaps remain, and both matter for mission-critical agentic systems.

Metric	Temporal	Akka
Availability SLA	99.99% (HA / multi-region)	99.9999%
Allowed downtime / year	~52 minutes	~31 seconds
RTO	~20 minutes (auto failover)	Sub-1 minute
RPO	Sub-1 minute	Zero byte
SLA scope	The orchestration layer	The entire platform

Temporal's SLA covers the orchestration layer. The memory, streaming, API, and governance services a customer bolts on operate under their own SLAs — or none.

The Determinism Tax

Temporal requires all workflow code to be **strictly deterministic** — no direct I/O, no unguarded randomness, no non-deterministic library calls — because recovery replays history. That constraint is at odds with agentic AI:

- LLM calls are non-deterministic by nature and must be carefully wrapped in activities.
- Any library with internal randomness must be audited or replaced.
- A non-determinism bug surfaces as a replay failure in production; debugging it needs deep Temporal expertise.

Akka provides durability and fault tolerance without imposing a determinism constraint on application code: state is event-sourced and the runtime handles recovery, so teams write natural business logic.

Cost

AI systems built with Akka are up to **90% cheaper to operate** than the equivalent assembled stack. Temporal bills **per action** — every step, signal, query, and timer is metered at $50 per million (stepping to $25 at high volume) — and that meter covers only orchestration; memory, streaming, APIs, observability, and governance are billed separately.

Akka runs all of it on one shared-compute runtime. The efficiency comes from actor concurrency (~10T tokens/core/year vs ~2T; ~80% less compute than Python frameworks; Manulife: up to 300% more concurrency, 30–50% faster processing after porting from Python), shared compute, and micro-checkpointing that minimizes retries. The spend is predictable — a fixed annual fee, not per-action metering that moves with load.

Continuous AI Intelligence

Akka Optimize keeps improving the AI after it ships. It runs an evaluation loop on your own production traffic, grading live interactions and training smaller specialized models on your proprietary data, so the system keeps getting better as it runs. The loop delivers three results that compound over time:

- **Model leverage** — route each request to the best model from any vendor under your policies, and reserve frontier models for the requests that need them.
- **Data sovereignty** — the specialized models are trained on your proprietary data, owned by you, and run inside your own environment.
- **Cost governance** — full visibility and control of AI spend, with savings that compound as the loop keeps running.

The evaluations run continuously inside Akka and grade traffic from agents wherever they run, in Akka or in third-party harnesses.

Governance and the EU AI Act

Temporal holds SOC 2 Type II, GDPR, and HIPAA for its infrastructure. Operators can pause, cancel, or terminate running work, and Workflow Pause is in pre-release. Each workflow keeps an append-only Event History, which Temporal calls an audit log and does not describe as tamper-evident. Temporal ships no AI-policy enforcement, decision explainability, or pre-deployment classification of its own. Partner guardrails such as Tenuo block an Activity before it runs, on Temporal workflows only.

The penalties are enforceable now

Violation	Maximum Fine
Prohibited AI practices (Art. 5)	EUR 35M or 7% global turnover
High-risk obligations (Art. 9-15)	EUR 15M or 3% global turnover

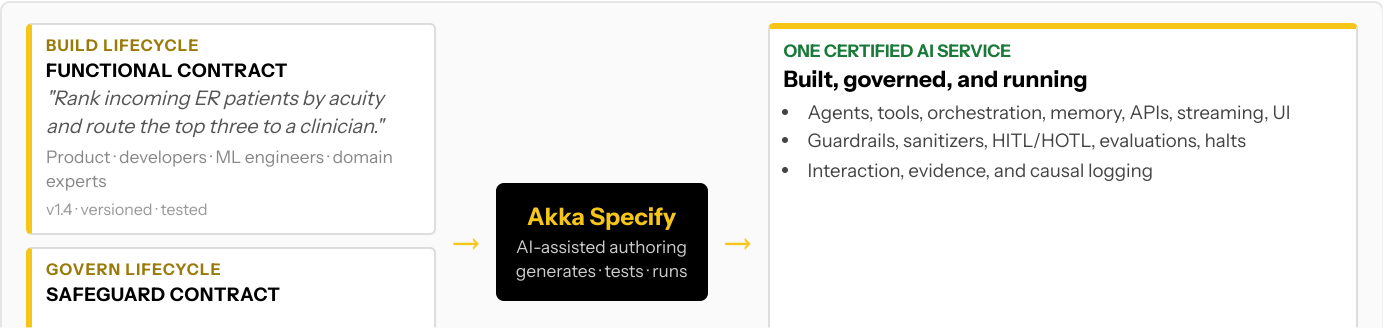
High-risk AI carries a 10-year logging-retention obligation (Art. 72).

How Akka governs

At the runtime: inline guardrails, policies, LLMs-as-a-judge, and sanitizers; hash-chained immutable evidence; HITL/HOTL control; classification against 241 regulations and frameworks and 1,718 controls before a system ships; multi-persona sign-offs; a sealed Governance Posture Package; and Akka Verify proving conformance from the running system. Governance Temporal would have to bolt on, Akka enforces inline.

Two Lifecycles, One Certified System

Building on Temporal means engineers writing deterministic workflow code; there is no path for a product manager, domain expert, or risk officer to contribute, and no built-in governance lifecycle. Akka runs two independent lifecycles on one platform via **Akka Specify**:



"Block prohibited practices under EU AI Act Article 5; notify regulators within 24h of any incident."

Risk · security · compliance

v2.1 · versioned & tested independent of the build

Akka Verify ↻ validates the running system against both specs and fine-tunes the AI from production data.

The build lifecycle and the governance lifecycle are versioned and tested independently, by different audiences — an audience and a workflow Temporal has no equivalent for.

Real-Time Streaming at Petabyte Scale

Temporal Workflow Streams, in public preview, streams agent output from a workflow to a user interface. High-throughput data pipelines are provisioned separately. Akka streaming is built into the runtime — continuous, backpressured, **petabyte-scale, in-memory**, with no external broker — powering both agent feedback loops and high-throughput data processing (the engine behind Tubi's real-time hyper-personalization at 5 billion tokens per second).

For the Buyer: Risk, Compliance, and Accountability

Buyer concern	Temporal	Akka
Certifications & audits	SOC 2 Type II, GDPR, HIPAA	19 standards — SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF — plus annual pen tests, SBOMs, 40+ policies (trust.akka.io)
Scope of accountability	The orchestration layer; you integrate and operate the rest	One platform, one SLA, 24/7 SRE — Akka owns the running system
Risk transfer	Standard cloud terms	Availability and data-integrity guarantees backed by contractual indemnities
Track record & funding model	Venture-funded: $550M Series E at a $12.55B valuation (September 2026), after a $300M Series D in February 2026; scaling on investor capital toward profitability	Profitable and self-funding; since 2009 and 100,000+ deployments (52 banks); Dell Technologies Capital is largest shareholder, a customer, and an AI partner
Budget predictability	Per-action metering that scales with load	Fixed annual fee finance can forecast

Both are well-capitalized — but Akka funds its roadmap and support from profit, not the next round, so the platform you standardize on does not depend on a venture timeline. The decision is scope and accountability: Temporal gives you a best-in-class orchestration layer to build a platform around; Akka gives you the platform.

Customers Running Agentic and Real-Time Systems on Akka

Manulife
2,000
developers across 100 projects on one governed platform

Tubi
5B tok/s
real-time hyper-personalization engine

Swiggy
71ms
order-assignment AI, ~50% latency reduction

John Deere
1,000+
tractor sensors turned into real-time insight

Verizon
750%
order-processing capacity gain; 6s → 2.4s response

Common Questions

We already run Temporal in production. Why add Akka?

Temporal is strong for durable workflow orchestration. If you are moving into agentic AI you also need native agents, durable memory, data streaming, and runtime governance, which Temporal leaves to third-party frameworks and partner integrations. Akka can run alongside existing Temporal workflows while you build the agentic layer on one platform.

Temporal raised $550M at a $12.55B valuation in September 2026 and targets agentic AI. Doesn't that close the gap?

Strong funding reflects real demand for durable execution. But Temporal's agentic story is an SDK integration on top of orchestration;

Can we add governance on top of Temporal?

Temporal supports governance through partner integrations. Tenuo checks authorization before each Activity runs, and Temporal can pause, cancel, or terminate running work. Temporal and partner controls act only on Temporal workflows and map to no named regulation. Temporal offers no pre-deployment classification. Akka Verify applies any control to any AI system, guarantees its enforcement, and covers pre-deployment governance.

Is Temporal cheaper because it is open source?

Temporal's core is open source, but production Temporal means Temporal Cloud (per-action billing) or self-hosting (you build and

Akka is built for agentic AI end to end — native agents, built-in memory, embedded governance, and a full-stack runtime. Capitalization does not add the missing layers.

operate HA/DR, observability, and everything else). The Akka shared-compute model is up to 90% cheaper to operate than the equivalent assembled stack, on a fixed annual fee.

Sources

Temporal Cloud SLA: docs.temporal.io/cloud/sla — 99.9% standard; 99.99% HA / multi-region
Temporal HA/DR: docs.temporal.io/cloud/high-availability, /rpo-rto — sub-1-min RPO, ~20-min RTO, automatic failover
Temporal Cloud pricing: docs.temporal.io/cloud/pricing — $50→$25 per million actions ($0.00005/action)
Temporal Series E: $550M at $12.55B valuation, Sep 14, 2026, led by Lightspeed; run-rate revenue above $250M; 4,300+ paying customers (temporal.io/news/temporal-raises-550m-at-a-12-55b-valuation); prior $300M Series D at $5B, Feb 2026
Temporal Workflow Streams: public preview, Jun 17, 2026 (temporal.io/blog/workflow-streams-live-interactivity-agents-other-applications)
Temporal governance: Workflow Pause, pre-release (docs.temporal.io/encyclopedia/workflow/workflow-pause); Event History (docs.temporal.io/workflow-execution/event); control-plane audit logs, 30 days (docs.temporal.io/cloud/audit-logging); partner guardrails (docs.temporal.io/develop/python/integrations/, tenuo.ai/temporal)
Temporal security: temporal.io/security — SOC 2 Type II, GDPR, HIPAA
Akka trust center: trust.akka.io — 19 compliance standards; SOC 2 II + public SOC 3; annual pen tests, SBOMs, 40+ policies
Akka performance: akka.io/blog/go-slow-to-go-fast — Manulife up to 300% more concurrency, 30–50% faster; ~10T vs ~2T tokens/core; ~80% less compute than Python
Akka platform: 99.9999% availability, active-active HA/DR, sub-1 min RTO, zero byte RPO (contractual indemnities); 241 regulations and frameworks / 1,718 controls; 100,000+ deployments / since 2009; profitable; Dell Technologies Capital